Deutsch-Jozsa Algorithm

The game...

- Input x is an N-bit binary string (N over 01001 0...)
- $f(x)$ maps to 0 or 1 (ie. N-bits → 1 bit)
- $f(x)$ → 'Constant' - eg. $f(x)=0$ no matter the input or $f(x)=1$ " " " "
  → 'Balanced' - outputs have an equal # 1s & 0s

eg.

x	$f(x)$
0	0
1	0

'Constant'

x	$f(x)$
0	1
1	0

'Balanced'

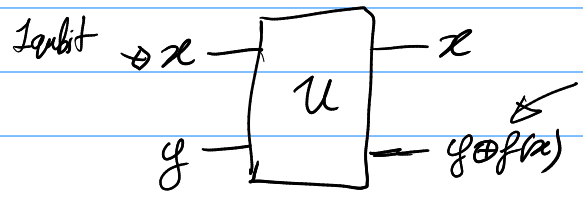
x	$f(x)$
00	0
01	1
10	1
11	0

Balanced.

Goal - find out if $f(x)$ is constant or balanced.
Classically this requires $\frac{2^N}{2}+1$ calls.

Quantum 1-bit case. (2 qubits)

A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

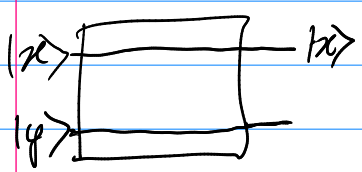


$f(x)$ – if constant

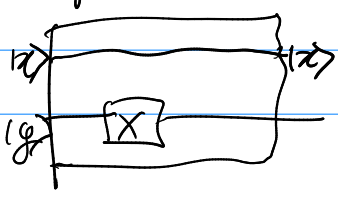
$f(x)$	y	$f(x)\oplus y$
0	$\|0\rangle$	$\|0\rangle$
1	$\|0\rangle$	$\|1\rangle$
0	$\|1\rangle$	$\|1\rangle$
1	$\|1\rangle$	$\|0\rangle$

Implementing the oracle U.

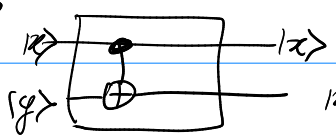
Constant $f(x)=0, \forall x$



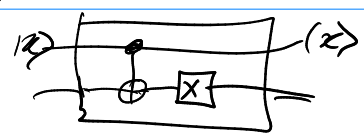
Constant $f(x)=1, \forall x$

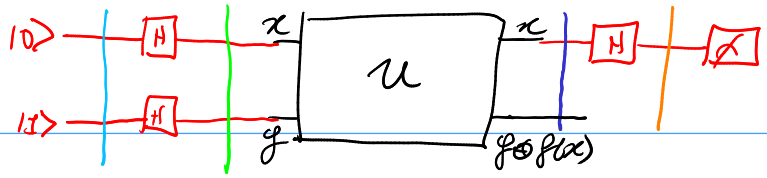


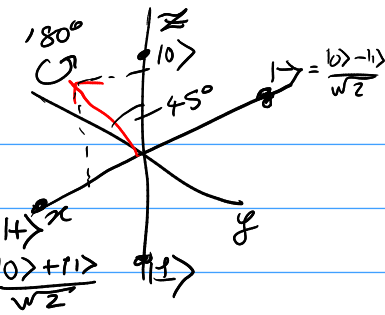
Balanced. $f(x)=x$



Balanced $f(x)=\overline{x}$ ($f(0)=1$, $f(1)=0$)







$|\psi_0\rangle = |0\,1\rangle$

$|\psi_1\rangle = |+\rangle\otimes|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)\otimes\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)$

$= \frac{1}{2}\cdot(|00\rangle - |01\rangle + |10\rangle - |11\rangle)$

$\forall x,\ f(x)=0 \qquad |\psi_2\rangle = \frac{1}{2}\cdot(|00\rangle - |01\rangle + |10\rangle - |11\rangle) = |+\rangle\otimes|-\rangle$

$\forall x,\ f(x)=1 \qquad |\psi_2\rangle = \frac{1}{2}\cdot(|01\rangle - |00\rangle + |11\rangle - |10\rangle) = -|+\rangle\otimes|-\rangle$

$\forall x,\ f(x)=x \qquad |\psi_2\rangle = \frac{1}{2}\cdot(\underbrace{|00\rangle}_{x=0\Rightarrow f(x)=0} - |01\rangle + \underbrace{|11\rangle}_{x=1\ f(x)=1} - |10\rangle) = |-\rangle\otimes|-\rangle$

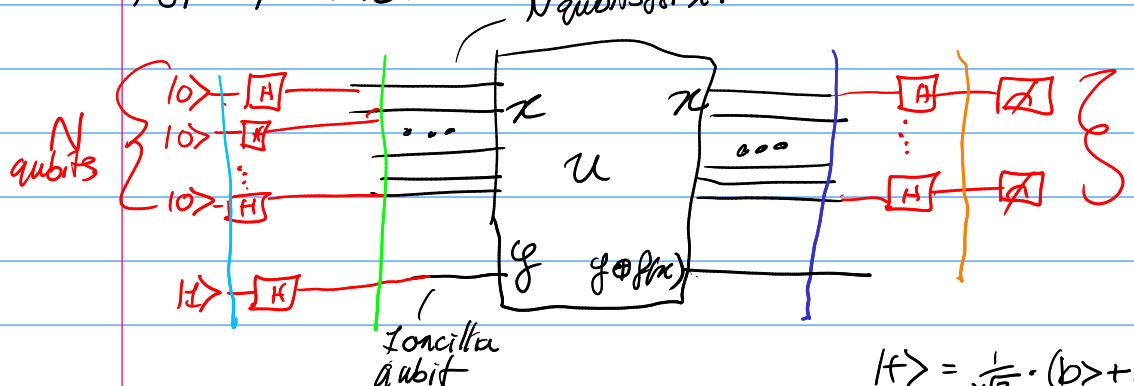
$\forall x \quad f(x)=\bar{x} \qquad |\psi_2\rangle = \frac{1}{2}\cdot(|01\rangle - |00\rangle + |10\rangle - |11\rangle) = -|-\rangle\otimes|-\rangle$

So constant $f(x)$ leaves first qubit on $|+\rangle$
balanced $f(x)$ " " " " $|-\rangle$

∴ output on 1st qubit is $|0\rangle$ – constant $f(x)$
$|1\rangle$ = balanced $f(x)$

Only requires 'one evaluation'.

For N-bits.



Check if all $|0\rangle$ if / check if it is $|0000\ldots0\rangle$

$|0\rangle^{\otimes N} = |0\ldots0\rangle$ (N)

$|\psi_0\rangle = |0\rangle^{\otimes N}\otimes|1\rangle$

$|\psi_1\rangle = |+\rangle^{\otimes N}\otimes|-\rangle$

$= \frac{1}{\sqrt{2^N}}\cdot\sum_{x\in\{0,1\}^N}|x\rangle\otimes|-\rangle$

$|\psi_2\rangle = \frac{1}{\sqrt{2^N}}\cdot\sum_{x\in\{0,1\}^N}(-1)^{f(x)}\cdot|x\rangle\otimes|-\rangle$

$|+\rangle = \frac{1}{\sqrt{2}}\cdot(|0\rangle+|1\rangle)$

$|+\rangle\otimes|+\rangle = \frac{1}{2}\cdot(|0\rangle+|1\rangle)\otimes(|0\rangle+|1\rangle)$

$= \frac{1}{2}\cdot(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

$|+\rangle\otimes|+\rangle\otimes|+\rangle = \frac{1}{\sqrt{2^3}}\cdot(|000\rangle + |001\rangle + \ldots)$

$f(x)=0 \Rightarrow g\oplus f(x) = |-\rangle$

$f(x)=1 \Rightarrow g\oplus f(x) \Rightarrow |0\rangle-|1\rangle \to |1\rangle-|0\rangle \equiv -|-\rangle$

Σ of all bit strings.

$|000000\rangle \cdot (-1)^{f(00000)}$

$|100001\rangle \cdot (-1)^{f(00001)}$

$|110010\rangle \cdot (-1)^{f(10010)}$

$|\psi_2\rangle \rightarrow$

coefficient is + if $f(x)=0$

− if $f(x)=1$

Hadamard

$\frac{1}{\sqrt{2^N}} |+\rangle \otimes |+\rangle \otimes |+\rangle \otimes |+\rangle \otimes |-\rangle \otimes |+\rangle$

$\Downarrow$

measure $|000000\rangle$

$|000000| \left(|000000\rangle \pm |000001\rangle \right)$

$\Downarrow$

$\frac{1}{\sqrt{2^N}}$

$\langle 00000 | H^{\otimes N} |x\rangle = \frac{1}{\sqrt{2^N}}$

$$P_{\text{all zeroes}} = \left(\frac{1}{\sqrt{2^N}} \cdot \sum_{x \in \{0,1\}^N} (-1)^{f(x)} \cdot \frac{1}{\sqrt{2^N}} \right)^2$$

$$= \left(\frac{1}{2^N} \cdot \sum_{x \in \{0,1\}^N} (-1)^{f(x)} \right)^2$$

If $\forall x, f(x)=0 \Rightarrow P_{\text{all zeroes}} = \left(\frac{2^N}{2^N}\right)^2 = 1$

$\forall x, f(x)=1 \Rightarrow P_{\text{all zeroes}} = \left(\frac{-2^N}{2^N}\right)^2 = 1$

$\forall x, f(x)$ is balanced $\Rightarrow P_{\text{all zeroes}} = 0$

Ex: If all zeroes $\Rightarrow$ constant

Otherwise $\Rightarrow$ balanced.

Gottesman Knill

Grover's Algorithm

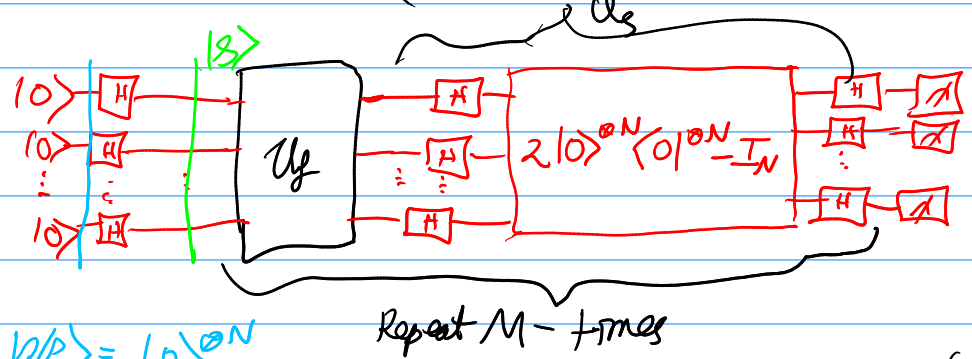
The game.

- Consider a lock with an N-bit password.
- Find this password q
- No information is given for a trial $x = \overbrace{01100\ldots100}^{N\text{bits}}$ other than pass/fail.

Classically we need up to 2^N trials/guesses required. For quantum case, we build another 'Oracle'

$$U_f|x\rangle = \begin{cases} |x\rangle & x \text{ is } \underline{\text{NOT}} \text{ the password} \\ -|x\rangle & x \text{ is the password } q. \end{cases}$$

($-1 = e^{i\pi}$ eg. / a CZ but controlled on $f(x)$ to check password)



$|\psi_0\rangle = |0\rangle^{\otimes N}$

$|\psi_1\rangle = \frac{1}{\sqrt{2^N}} \sum_{x \in \{0,1\}^N} |x\rangle = |s\rangle$

$2|0\rangle\langle 0| - I_2$

$2\begin{pmatrix}1\\0\end{pmatrix}(1\ 0) - \begin{pmatrix}1&0\\0&1\end{pmatrix}$

$= \begin{pmatrix}2&0\\0&0\end{pmatrix} - \begin{pmatrix}1&0\\0&1\end{pmatrix}$

$= \begin{pmatrix}1&0\\0&-1\end{pmatrix} = \sigma_z$

Let's first try to unpack the Grover Diffusion Operator

$$U_s = 2|s\rangle\langle s| - I_N$$

But what does it do?

$|s'\rangle$ is everything $\perp$ to $|s\rangle$ i.e. orthogonal.

$$U_s(\alpha|s\rangle + \beta|s'\rangle)$$

α in $|s\rangle$; β not in $|s\rangle$

arbitrary vector.

$$U_s(\alpha|s\rangle + \beta|s'\rangle) = (2|s\rangle\langle s| - I_N)\cdot(\alpha|s\rangle + \beta|s'\rangle)$$

$$= 2\alpha|s\rangle - \alpha|s\rangle + 2\beta|s\rangle\langle s|s'\rangle - \beta|s'\rangle$$

$\langle s|s'\rangle = 0$ ∴ orthogonal

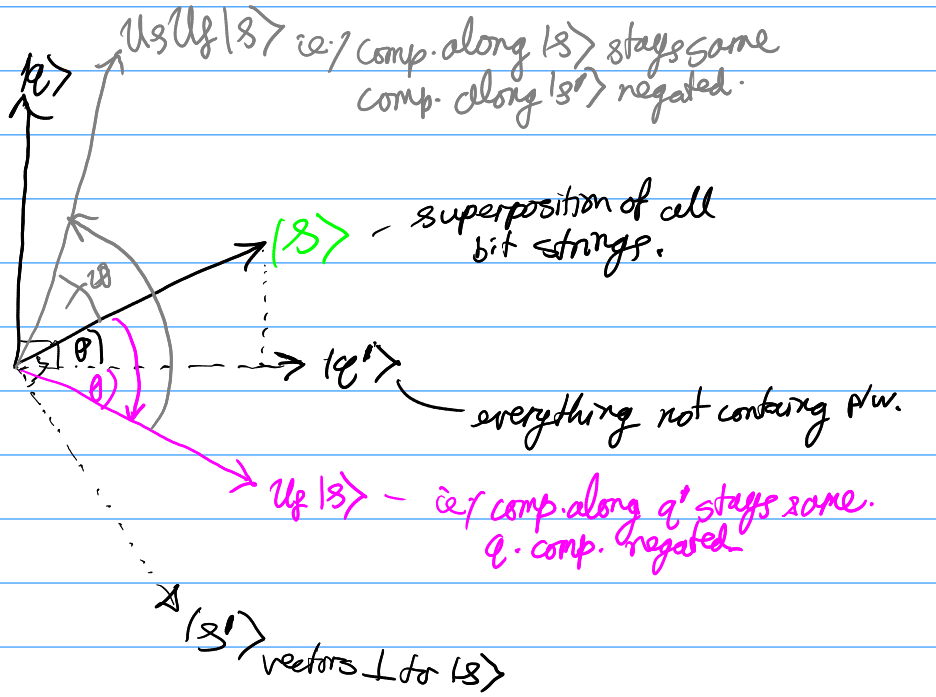
$$= \alpha|s\rangle - \beta|s'\rangle$$ - i.e. basically the same thing but checks in a (-) sign on the orthogonal components!

What does U_f look like?

$$U_f = I_N - |q\rangle\langle q| - |q\rangle\langle q| = I_N - 2|q\rangle\langle q|$$

- I_N: keep everything
- $|q\rangle\langle q|$: remove q
- $-|q\rangle\langle q|$: put it back w/ negative sign

i.e. same as input vector but (-) sign on $|q\rangle$ component.



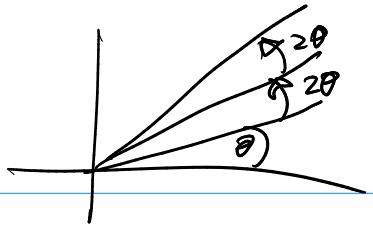
⇒ So each $U_f U_s$ pass gets it 2θ towards the goal.

To find # pass required...

$\sin(\theta)$ = projection of $|s\rangle$ (all possible 2^N bit strings) onto correct password $|q\rangle \equiv \frac{1}{\sqrt{2^N}}$

θ is small, $\Rightarrow \sin(\theta) \approx \theta = \frac{1}{\sqrt{2^N}}$

I exaggerated in diagram above



ie: $2\theta \cdot M + \theta = \pi/2$

M iterations

$\therefore 2M = \frac{\pi}{2\theta} - 1$

$\therefore M = \frac{\pi}{4\theta} - \frac{1}{2}$

$\therefore M \sim \frac{\pi}{4} \cdot \sqrt{2^N}$

ie: classically we require 2^N trials.
here we need $\sim \sqrt{2^N}$

ie: $O(\sqrt{N})$ - ie: quadratic speedup.

WIFI.

password 123 → [hashing] → 0x AB130....

is my hash matching the correct hash.

0x CD 123...